Foundation models for marketplace prediction

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9.4% lower fare MAE than tuned boosting - zero domain training

Six-route public NYC pilot; not a comparison with Uber's production stack.

The opportunity. Foundation forecasts could complement Uber's existing prediction stack; economic structure can connect predictions to decisions. Chronos-2 is a time-series foundation model, extending my broader work on foundation models for structured and tabular data.

What I built. A reproducible local test, reusing my GARP-EFM infrastructure on an Apple M1: processed 62.9 million public NYC trip records and retained roughly 237,000 eligible rides across six short routes. Chronological training/validation preceded a March test of 1,574 common route-hour-horizon cases against six baselines.

The evidence. Zero-shot Chronos-2 with calendar and past trip counts achieved **$1.067 MAE versus $1.177 for gradient boosting: 9.4% lower error**. This is realized rider-fare forecasting, not upfront quotes, identified surge, or a comparison with Uber's production models. Tail errors and interval calibration still need work. A 60-step synthetic LoRA pilot gave no material gain; validation retained zero-shot.

A research hypothesis. A marketplace-specific synthetic curriculum could improve forecasting of arrivals, availability and disruption dynamics. In a separate toy prototype, joint zero-shot arrival/capacity forecasts were nearly tied with trained boosting. An economic calculator then maps candidate prices to ETA and conversion; its unsafe constraint failures show why shadow evaluation and guardrails come first.

Business case: a scenario, not measured lift. At 20% eligibility, 60% baseline conversion, 10% incremental-volume contribution, $1 million annual program cost and unchanged average fare, a **net causal +0.05 percentage-point conversion lift** (5 extra completions per 10,000 quotes) implies $1.93 million contribution and $0.93 million net annually. Scale: $115.952 billion, four times Q2 2026 Mobility Gross Bookings, not a forecast. No lift is inferred from the MAE result.

Proposed next step. With internal quotes, nonrequests, availability and ETA data, shadow a prediction service against the current stack. Establish calibration, tail performance and serving reliability jointly with engineering before a constrained experiment. The decision is whether this merits an internal pilot - not whether to replace pricing today.

Reproducible code, results and technical appendix